MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth Semester
Discipline Specific Core Course (DSC) - MG4DSCPEG200 –
ATF AND LATEST DEVELOPMENT IN FUEL TECHNOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

(Students should attempt at least one question from each course outcome to enhance their overall outcome attainability)

Part A

Multiple choice questions

Answer **all** questions

Each question carries **1** mark

1.Identify the primary objective of integrating biofuel production into biorefineries. [U]/[CO1]
a) Maximum utilization of biomass
2.Write the suitable technology when multiple co-products (corn oil, gluten meal) are required. [A]/[CO2]
b) Wet milling
3.Compare ethanol and gasoline based on calorific value. [U]/[CO2]
b) Ethanol has lower calorific value than gasoline
4.The primary purpose of filtration of aviation fuel before refuelling is to [K]/[CO3]
b) Remove suspended solids and rust particles
5. Which of the following aviation fuels has a high flash point and low volatility? [U]/[CO4]
c) Aviation kerosene (Jet fuel)
6.In the ATJ process, aviation fuel is produced from: [K]/[CO4]
c) Alcohols derived from biomass
7.Motor Octane Number (MON) is measured under[K]/[CO4]
b) Severe operating conditions
8.Low electrical conductivity of ATF may lead to: [K]/[CO5]
b) Static electricity accumulation
9. Excess aromatic content in ATF leads to: [U]/[CO5]
a) Increased smoke formation
10. ASTM D7566 is related to: [K]/[CO6]
b) Certification of aviation biofuels

[1x10= 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **3** marks

11. Explain the importance of waste utilization in biofuel production. [U]/[CO1]

Waste utilization in biofuel production is important because it helps in the efficient use of organic waste materials such as agricultural residues, food waste, and industrial by-products,

converting them into valuable fuels. This reduces environmental pollution and waste disposal problems while minimizing dependence on fossil fuels. Additionally, it enhances sustainability by lowering production costs and promoting a circular economy, where waste is transformed into useful energy resources. (Any relevant points give full credits)

12. Define lubricity and explain why lubricity additives are needed in modern jet fuels. [K]/[CO4]

Lubricity is the ability of a fuel to provide adequate lubrication between moving metal surfaces, reducing friction and wear in fuel system components such as pumps and injectors.

In modern jet fuels, lubricity additives are needed because highly refined fuels (with low sulfur and aromatic content) naturally have poor lubricating properties. This can lead to increased wear, scuffing, and damage to fuel system parts. Therefore, lubricity additives are added to improve the fuel's ability to protect components, ensuring smooth operation, reduced maintenance, and longer service life of the engine system. (definition 1 mark and Two points explanation 2 marks)

13. Define aviation fuels and state their main classification based on engine type. [K]/[CO3]

Aviation fuels are specialized fuels used to power aircraft engines, designed to meet strict requirements such as high energy content, safety, stability, and reliable performance under varying altitude and temperature conditions.

Based on engine type, aviation fuels are mainly classified into:

1. Aviation gasoline (AvGas) – used in piston (reciprocating) engines.
2. Aviation turbine fuel (ATF or jet fuel) – used in jet engines such as turbojets, turbofans, and turboprops (Definition 1 mark and classification 2 marks)

14. Discuss on total additive sulphur in aviation fuels. [U]/[CO4]

Total additive sulphur in aviation fuels refers to the sulphur content present in the fuel due to naturally occurring sulphur compounds and, in some cases, from additive components. In aviation turbine fuels, sulphur is not intentionally added for performance; instead, its level is strictly controlled within specified limits. The presence of sulphur is important because excessive sulphur can lead to corrosion of engine parts and formation of harmful combustion products such as sulphur oxides, which contribute to environmental pollution. However, very low sulphur content may reduce natural lubricity of the fuel. Therefore, maintaining an optimum, limited sulphur content is essential to balance engine protection, fuel performance, and environmental considerations. (Any 3 points give 3 marks)

15. State any three chemical properties of Aviation Turbine Fuel. [K]/[CO5]

Aviation Turbine Fuel (ATF) exhibits important chemical properties that ensure safe and efficient performance in aircraft engines. It contains very low sulphur content, which minimizes corrosion and reduces sulphur oxide emissions. ATF has good thermal and oxidation stability, meaning it resists decomposition, oxidation, and the formation of gums or sludge during storage and high-temperature operation. Additionally, it is non-corrosive in nature with low acidity, ensuring protection of fuel system components and engine parts. (Any 3 points give 3 marks)

16. Examine the limitations of hydrogen as an aviation fuel. [An]/[CO6]

Hydrogen has potential as a clean aviation fuel, but it has several important limitations. One major issue is its low volumetric energy density, which means it occupies a much larger volume than conventional jet fuel, requiring bulky and specially designed storage tanks. It must be

stored either at very high pressures or in liquid form at extremely low temperatures (cryogenic conditions), making storage and handling complex and costly. Hydrogen also has high flammability and requires strict safety measures to prevent leakage and fire hazards. In addition, the current aviation infrastructure is not compatible with hydrogen, requiring major modifications to aircraft design, fuel systems, and refuelling facilities. Finally, large-scale production of hydrogen, especially green hydrogen, is still expensive and energy-intensive, limiting its widespread adoption in aviation. (Any 3 points give full credits)

[3x4= 12]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **7** marks

17. Outline anaerobic digestion for biogas production. [K]/[CO1]

Anaerobic digestion is a biological process in which organic matter such as animal dung, agricultural waste, and food waste is decomposed by microorganisms in the absence of oxygen to produce biogas. The process takes place in an airtight digester and results in the formation of biogas (mainly methane and carbon dioxide) and a nutrient-rich residue called digestate.

The process occurs in four main stages. Hydrolysis is the first step, where complex organic materials like carbohydrates, proteins, and fats are broken down into simpler soluble compounds. This is followed by acidogenesis, in which these products are converted into volatile fatty acids, alcohols, hydrogen, and carbon dioxide by acid-forming bacteria. The third stage, acetogenesis, converts these intermediate products into acetic acid, hydrogen, and carbon dioxide. Finally, in methanogenesis, methanogenic bacteria convert acetic acid and hydrogen into methane and carbon dioxide, which form the main components of biogas.

Anaerobic digestion is carried out under controlled conditions of temperature, pH, and moisture to ensure efficient microbial activity. The biogas produced is used as a renewable fuel for cooking, heating, and electricity generation, while the digestate is used as an organic fertilizer. Thus, anaerobic digestion is an eco-friendly method for waste management and sustainable energy production. (Any seven required points give 7 marks)

18. Illustrate a suitable production technology for an ethanol plant designed for low capital investment and simpler operation. [A]/[CO2]

A suitable production technology for an ethanol plant with low capital investment and simple operation is the dry milling process. In this method, the entire grain (such as corn) is used without separating its components, making the process economical and less complex compared to wet milling.

In the dry milling process, the raw material is first cleaned and ground into flour (meal). The ground material is then mixed with water to form a slurry, followed by liquefaction, where enzymes (like α-amylase) break down starch into shorter chains. This is followed by saccharification, where enzymes such as glucoamylase convert starch into fermentable sugars (glucose). The resulting mash undergoes fermentation using yeast (Saccharomyces cerevisiae), producing ethanol and carbon dioxide. After fermentation, the mixture is subjected to distillation to separate ethanol, and further dehydration is carried out to obtain fuel-grade ethanol.

The advantages of this technology include low initial investment, simpler equipment, ease of operation, and suitability for small- to medium-scale plants. Although it produces fewer co-products compared to wet milling, it is widely preferred where cost and operational simplicity are important. (Any 5 relevant points give full credit)

19. Identify the advantages of AV gas fuel over turbine engine fuel. [K]/[CO3]

Aviation gasoline (AvGas) offers several advantages over turbine engine fuel (ATF) when used in piston-engine aircraft. It has a higher-octane number, which provides excellent anti-knock properties and ensures smooth combustion in high-compression piston engines. AvGas also has better volatility, allowing easy vaporization and reliable engine starting, especially under cold conditions. It ensures uniform air–fuel mixing, leading to efficient combustion and stable engine performance. Additionally, AvGas produces cleaner combustion with fewer deposits in piston engines compared to heavier fuels. It is well-suited for lightweight aircraft engines, offering good throttle response and operational simplicity. Overall, AvGas is specifically optimized for piston engines, providing reliability, efficiency, and consistent performance under varying flight conditions. (Any 5 points give full credit)

20. List the advantages and limitations of using SAF in modern aviation. [K]/[CO3]

Sustainable Aviation Fuel (SAF) offers several advantages in modern aviation. It significantly reduces greenhouse gas emissions compared to conventional jet fuel, helping to mitigate climate change. SAF is produced from renewable resources such as biomass, waste oils, and agricultural residues, promoting sustainability and reducing dependence on fossil fuels. It is compatible with existing aircraft engines and infrastructure (drop-in fuel), requiring minimal modifications. SAF also contains lower sulphur and aromatic content, resulting in cleaner combustion, reduced particulate emissions, and improved air quality. Additionally, it supports waste utilization and contributes to a circular economy.

However, SAF also has certain limitations. The high production cost makes it less economically competitive than conventional fuels. There is limited large-scale availability due to constraints in feedstock supply and production capacity. Some SAF pathways may face feedstock sustainability issues and competition with food resources. The production processes can be energy-intensive, and achieving consistent fuel quality can be challenging. Furthermore, widespread adoption requires policy support, infrastructure development, and technological advancements. Thus, while SAF is environmentally beneficial, economic and scalability challenges limit its current usage.

(Any 3 points from advantages and limitations give 3.5 marks each)

21. State the hydrotreating process used in the manufacture of ATF, including operating conditions, reactions involved, and improvements in fuel quality. [K]/[CO5]

Hydrotreating is a key refinery process used in the manufacture of aviation turbine fuel (ATF) to improve its quality and ensure it meets strict aviation specifications. In this process, the kerosene fraction obtained from crude oil distillation is mixed with hydrogen and passed over a catalyst (commonly cobalt–molybdenum or nickel–molybdenum on alumina) at high temperature and pressure. The hydrogen reacts with impurities such as sulphur, nitrogen, oxygen compounds, and unsaturated hydrocarbons, converting them into stable products like hydrogen sulphide and ammonia, which are then removed. Hydrotreating improves the fuel's thermal stability, reduces corrosiveness, lowers sulphur content, and enhances combustion characteristics. After the reaction, the treated fuel is cooled, gases are separated, and the

purified ATF is recovered, meeting the required standards for safe and efficient aircraft operation. (statement about hydrotreating 2 marks, operation condition 1 mark, reactions 2 marks and fuel quality improvement 2 marks)

22. Describe the hydrodesulphurization process used in the production of Aviation Turbine Fuel, highlighting its role in improving fuel quality. [U]/[CO5]

The kerosene fraction obtained from atmospheric distillation is treated with hydrogen in the presence of a suitable catalyst in a process known as hydrotreating. During this process, sulphur compounds are converted into hydrogen sulphide (H_2S) and removed, while nitrogen and oxygenated compounds are also eliminated. This purification step significantly enhances the quality of the fuel by improving its thermal stability, which reduces gum and deposit formation during high-temperature operation in turbine engines. Hydrotreating also increases the smoke point, ensuring cleaner combustion, and improves the odour and colour of the fuel, making it suitable for use as aviation turbine fuel (ATF) in accordance with aviation fuel specifications.

In some refineries, the Merox process is employed to oxidize and remove mercaptans, reducing fuel corrosiveness and improving odour. (Process explanation 4 marks quality improvement 3 marks)

[7x4 = 28]